



Bringing water to life®

PROCTORS CREEK WWTP, VA
REPLACEMENT BOSKER SYSTEM
TROLLEY JUNCTION BOX

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DRAWN BY CHECKED BY DATE PROJECT REV	AD P 04/21/26 RSW1596-126 1	PROCTORS CREEK WWTP, VA REPLACEMENT BOSKER SYSTEM TROLLEY JUNCTION BOX	
D	SHEET 1 OF 6	RSW1596-126	A

INITIAL RELEASE	026455	AB	AC	04/21/26	DATE
REVISION	EN/ECO	BY	CHECKED	DATE	

CONSTRUCTION NOTES

CONTROL PANEL SPECIFICATIONS

COMPONENTS SPECIFIC

- 1. ENCLOSURE TO BE NEMA 4X, POLYCARBONATE CONSTRUCTION ENCLOSURE.
- 1.1 ATTACH NAMEPLATES WITH MANUFACTURER'S RECOMMENDED ADHESIVE TO MAINTAIN NEMA STANDARD OF ENCLOSURE
- 1.2 CONTROL PANEL SHALL BE LABELED WITH SERIALIZED UL LABEL
- 1.3 PANEL WILL BE LABELED WITH IDENTIFICATION PLATE AS NOTED ON ENCLOSURE DRAWING.
- 1.4 ALL RELAY CONTACTS ARE RATED AT 10 AMPS UNLESS OTHERWISE NOTED BY THE PART MANUFACTURER.
- 1.5 COMPONENT IMAGES MAY NOT ACCURATELY REPRESENT THE ACTUAL DEVICE IN SOME INSTANCES. DIMENSIONS OF DEVICES, HOWEVER, WILL BE ACCURATE FOR PURPOSES OF LAYOUT AND SPACING.
- 1.6 SUBSTITUTIONS FOR COMPONENTS MAY BE PERMITTED UPON APPROVAL BY OVIVO ENGINEERS.
- 1.7 PANEL ENCLOSURE SHOULD BE POWDER COATED WHITE FINISHED.

WIRE SIZING & TYPE

- WIRE SIZING SHALL BE NO LESS THAN AS FOLLOWS (UNLESS OTHERWISE NOTED):
- 480 3 PHASE - 12 AWG TYPE: XHHW
- 120 1 PHASE - 14 AWG TYPE: XHHW
- 24V DC - 14 AWG TYPE: XHHW

WIRE COLOR

- WIRE COLOR SHALL BE AS FOLLOWS:
- POWER (480 VOLTS)
- GREEN - GROUND
- BROWN, ORANGE, YELLOW - PHASE
- CONTROL (120 VAC)
- RED - LOAD WIRES FOR ALL LOCAL CONTROL CIRCUITS
- WHITE - NEUTRAL FOR ALL LOCAL CONTROL CIRCUITS
- YELLOW - FOREIGN POWER IN PANEL
- WHITE W/ YELLOW STRIPE - NEUTRAL FOR ALL FOREIGN POWER IN PANEL
- CONTROL (24 VDC)
- BLUE - LOAD WIRES FOR ALL LOCAL CONTROL CIRCUITS
- WHITE W/BLUE STRIPE - COMMON FOR ALL LOCAL CONTROL CIRCUITS

CONNECTIONS

- 4. ALL CONNECTIONS SHALL BE POINT TO POINT WITHOUT SPLICES. EXCEPTIONS APPLY TO SOLENOIDS AND MOTORS AND FACTORY WIRED COMPONENTS.
- 4.1 ALL CONDUCTORS MUST BE MARKED WITH MACHINE PRINTED TAGS AT EACH TERMINATION. MARKERS SHALL BE OF HEAT SHRINK TYPE.

DOCUMENTATION

- 5. FABRICATOR SHALL PROVIDE ONE COPY OF ALL DOCUMENTS SUPPLIED WITH THE COMPONENTS SUPPLIED BY THE FABRICATOR. THIS INFORMATION INCLUDES BUT IS NOT LIMITED TO MANUFACTURERS SPECIFICATIONS, INSTALLATION MANUALS AND OPERATION MANUALS.
- 5.1 ANY OVIVO APPROVED CHANGES OR MARK-UPS TO THE SCHEMATICS, PANEL LAYOUTS AND ANY OTHER PERTAINING DOCUMENTS SHALL BE MAINTAINED AND DELIVERED TO OVIVO AT ACCEPTANCE OF THE PANEL.

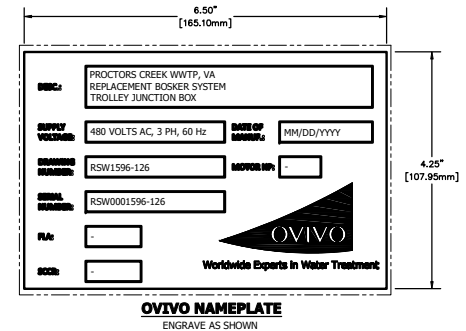
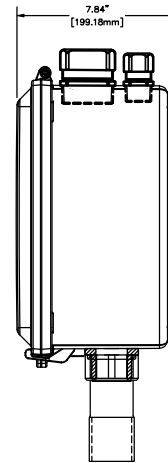
TESTING

- 6. OVIVO ENGINEERS OR OVIVO APPROVED FABRICATOR, WILL PERFORM A POINT TO POINT CONTINUITY TEST ON THE PANEL.
- 6.1 WHEN APPLICABLE, THE FABRICATOR WILL ALSO PROVIDE POWER TO THE 120VAC OR 480VAC CIRCUITS FOR VERIFICATION AND TESTING. FABRICATOR SHALL ALLOW A MINIMUM OF 2 HOUR OF ACCESS TO THE CONTROL PANEL BY ENGINEER FOR THIS AND PREVIOUSLY STATED TEST. IF POSSIBLE 3 PHASE POWER WILL BE PROVIDED FOR TESTING.

WIRING CONNECTIONS

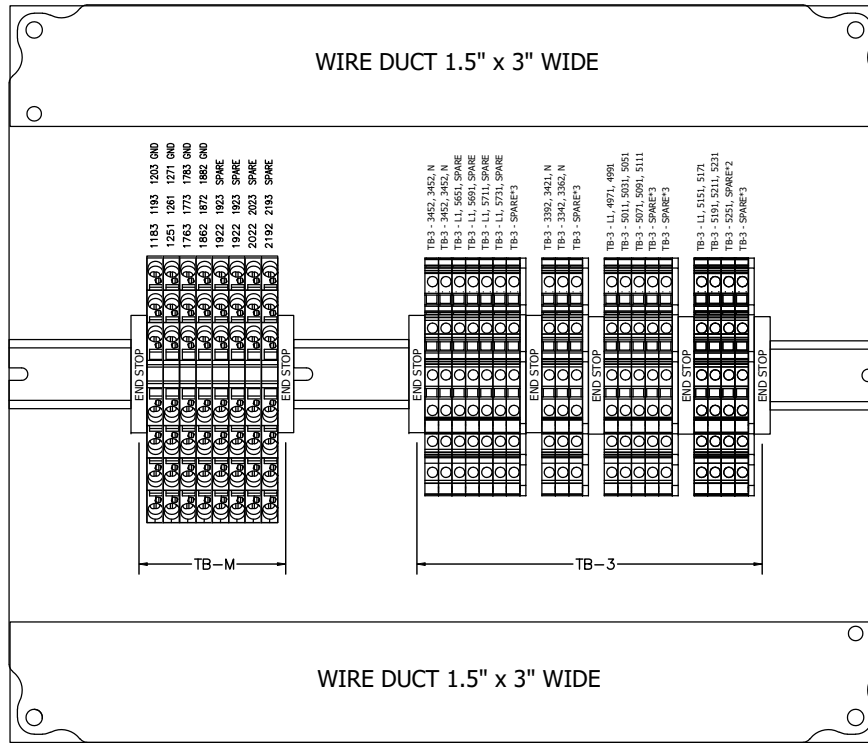
- 7.1 INTERNAL PANEL WIRING
- 7.2 FIELD WIRING (NOT OVIVO SCOPE)
- 7.3 FIELD WIRING (PROVIDED BY OVIVO, INSTALLATION BY OTHERS)
- 7.4 COMMUNICATION

GENERAL DRAWING NOTES



ITEM	QTY	TAGS	CATALOG	MAKE	DESCRIPTION
1	1	ENCLOSURE	SCE-1412PCW	SCE	POLYCARBONATE WALL-MOUNT JUNCTION BOX WITH WINDOW, NEMA 4X, 15.40"X9.469"X3.86" IN
2	1	ENCLOSURE	SCE-14P13AL	SCE	SUBPANEL BENT, FLAT ALUMINUM
3	8	TB-M	308838	PHENIX CONTACT	MOTOR TERMINAL BLOCK, 4-LEVEL, SPRING-CAGE CONNECTION, AWG 28-10, GRAY COLOR, 32A, 600 VOLTS
4	20	TB-3	3214259	PHENIX CONTACT	3-LEVEL TERMINAL BLOCK, SCREW CONNECTION, CROSS SECTION 0.14MM-4MM, GRAY COLOR, 19A, 500VOLTS
5	4	TB-3	3214914	PHENIX CONTACT	END COVER, GRAY COLOR
6	7	AS SHOWN	302226	PHENIX CONTACT	QUICK MOUNTING END CLAMP, DIN RAIL MOUNT, WITH PARKING OPTION AND PARKING OPTION, GRAY COLOR
7	AS REQUIRED	AS SHOWN	0807012	PHENIX CONTACT	DIN RAIL, NS 35/7.5, PERFORATED, 1 M
8	AS REQUIRED	AS SHOWN	TYD15X3WP4W WITH COVER TYD2CW6	THOMAS & BETTS BY ABB	WIDE SLOT WIRING DUCT, PVC, 1.5 INCHES X 3 INCHES, WHITE WITH COVER
9	3	AS SHOWN	H5-CG2	DUCT-O-WIRE	HEAT SHRINKABLE CABLE CONNECTORS, CGB, HEAT SHRINK, WITH SPARE
10	5	ACCESSORY	SEOM25G	HUBBELL	KELLEMS WIRE MANAGEMENT, CORD CONNECTORS, 0.49-0.74", M25 GRAY
11	1	ACCESSORY	SEOM40G	HUBBELL	KELLEMS WIRE MANAGEMENT, CORD CONNECTORS, 0.87-1.3", M40 GRAY
12	AS REQUIRED	ACCESSORY	LNVM25BK100	HUBBELL	NYLON LOCKNUT 25MM
13	AS REQUIRED	ACCESSORY	LNVM40BK100	HUBBELL	NYLON LOCKNUT 40MM

NOTE: INSTALLER TO BOND ENCLOSURE
TO STRUCTURE



CABLE DETAILS

GLAND #	GLAND TRADE SIZE	DESCRIPTION	CABLE SIZE	GLAND QTY
CONTROL SIDE CABLE SIZE				
1	HS-CG2	JB 1 TO JB 1A FOR GRIPPER SOLENOID (VIA FESTOON)	3#14	-
2		JB 1 TO JB 1A FOR WARNING HORN AND ALARM (VIA FESTOON)	3#14	-
3		JB 1 TO JB 1A FOR PROXIMITY SWITCH (VIA FESTOON)	8#14	-
4		JB 1 TO JB 1A FOR Limit SWITCH (VIA FESTOON)	7#14	-
5	M40	JB 1 TO FIELD JB (2 TO 5) FOR CONTROL SIGNALS	30#14, 1#12G	1
POWER SIDE CABLE SIZE VIA FESTOON				
6	HS-CG2	JB 1 TO JB 1A FOR TROLLY MOTOR 1 & BRAKE	6#14, 1#14G	-
7		JB 1 TO JB 1A FOR TROLLY MOTOR 2 & BRAKE	6#14, 1#14G	-
8		JB 1 TO JB 1A FOR GRIPPER MOTOR & BRAKE	6#14, 1#14G	-
9		JB 1 TO JB 1A FOR HYDRAULIC PUMP MOTOR & HYDRAULIC FLUID HEATER	6#14, 1#14G	-
POWER SIDE CABLE SIZE				
10	M25	JB 1 TO TROLLY MOTOR 1 & BRAKE	6#14, 1#14G	1
11	M25	JB 1 TO TROLLY MOTOR 2 & BRAKE	6#14, 1#14G	1
12	M25	JB 1 TO GRIPPER MOTOR & BRAKE	6#14, 1#14G	1
13	M25	JB 1 TO HYDRAULIC PUMP MOTOR & HYDRAULIC FLUID HEATER	6#14, 1#14G	1
POWER & CONTROL SIDE CABLE SIZE VIA FESTOON				
14	HS-CG2	JB 1 TO JB 1A FOR TROLLY MOTOR 1 & 2 HEATER	4#14	-
15		JB 1 TO JB 1A FOR GRIPPER & HYDRAULIC PUMP MOTOR HEATER	4#14	-
16		JB 1 TO JB 1A FOR MOTOR HEATERS THERMOSTAT	8#14	-
POWER SIDE CABLE SIZE				
17	M25	JB 1 TO TROLLY MOTOR 1 & 2 HEATER	4#14	1
18		JB 1 TO GRIPPER & HYDRAULIC PUMP MOTOR HEATER	4#14	

